

# Code-Layout, Readability and Reusability

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|---------------|---------------------------------------------------|
| Date          | 14 July 2026                                      |
| Team ID       |                                                   |
| Project Name  | HDI Predictor (Human Development Index Predictor) |
| Team Size     | 1 (Individual Project)                            |
| Maximum Marks | 5 Marks                                           |

## Code Layout Checklist

| S.No | Code Quality Parameter    | Description                                                       | Followed | Remarks |
|------|---------------------------|-------------------------------------------------------------------|----------|---------|
| 1    | Consistent Indentation    | Uniform 4-space indentation via PEP8 throughout                   | Yes      | —       |
| 2    | Proper File Structure     | Split into config.py, model_utils.py, db_utils.py, app.py         | Yes      | —       |
| 3    | Meaningful Variable Names | e.g. life_exp, expected_schooling, predicted_tier                 | Yes      | —       |
| 4    | Function / Method Names   | e.g. predict_hdi(), classify_tier(), get_history()                | Yes      | —       |
| 5    | Code Comments             | Docstrings on every module + inline comments on non-obvious logic | Yes      | —       |
| 6    | Modular Design            | Model, DB, and config each isolated from app.py routes            | Yes      | —       |
| 7    | No Redundant Code         | Shared helpers (e.g. classify_tier) used from one place only      | Yes      | —       |
| 8    | Error Handling            | try/except around DB ops; input range validation in /api/predict  | Yes      | —       |

## Reusable Components / Modules

| S.No | Component / Module Name            | Language / Tech    | Where Reused                                                  | Reusability |
|------|------------------------------------|--------------------|---------------------------------------------------------------|-------------|
| 1    | model_utils.predict_hdi()          | Python             | Called from /api/predict and seed_history.py                  | High        |
| 2    | db_utils (get/insert/delete/clear) | Python (SQLite)    | Used by every history-related route                           | High        |
| 3    | config.py constants                | Python             | Shared by app.py, model_utils.py, db_utils.py, train_model.py | High        |
| 4    | doc_builder.build_pdf()            | Python (reportlab) | Reused across all 22 deliverable PDFs in this repo            | High        |